



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 3 Intracellular Accumulation

By: Assistant Lecturer, Soroor M. Hadi

Learning Objectives

- ☐ **Know what intracellular accumulation is.**
- ☐ **Know the four pathways of intracellular accumulation.**
- ☐ **Know the categories of intracellular accumulation and describe its types and how they appear.**

Introduction

Intracellular accumulations are the buildup of abnormal amounts of various substances within the **cell**, which may be located in the **cytoplasm**, within **organelles (like lysosomes)**, or in the **nucleus**.

Location: Cytoplasm, organelles (lysosomes), or nucleus.

Source: The substance may be synthesized by the affected cell (**endogenous**) or be produced elsewhere and taken up by the cell (**exogenous**).

Four Main Pathways of Accumulation

1-Inadequate Removal of a Normal Endogenous Substance:

The rate of production is normal or increased, but the rate of catabolism (metabolism) or excretion is insufficient.

Example: Fatty change (steatosis) in the liver.

2-Accumulation of an Abnormal Endogenous Substance:

Due to genetic or acquired defects in protein folding, packaging, transport, or secretion.

Example: Accumulation of mutated alpha-1 antitrypsin in the liver (A1AT deficiency).

Four Main Pathways of Accumulation

3-The cell takes up a substance (endogenous or exogenous) that is non-degradable or is an indigestible residue of a prior process.

Example: such as lipofuscin and hemosiderin.

4-Deposition and Accumulation of an Abnormal Exogenous Substance:
The cell has no enzymatic machinery to degrade or transport the substance.

Example: Carbon (coal dust) particles in the lungs.

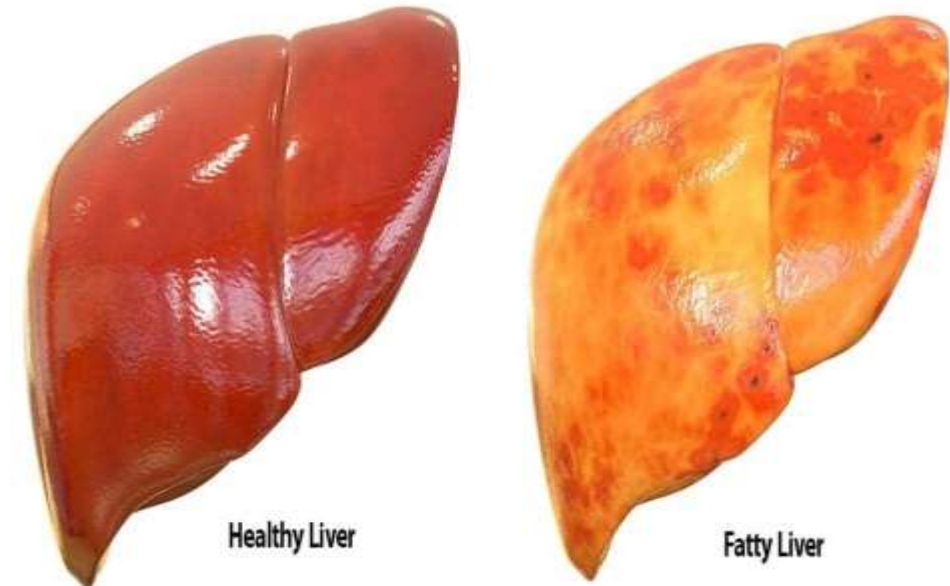
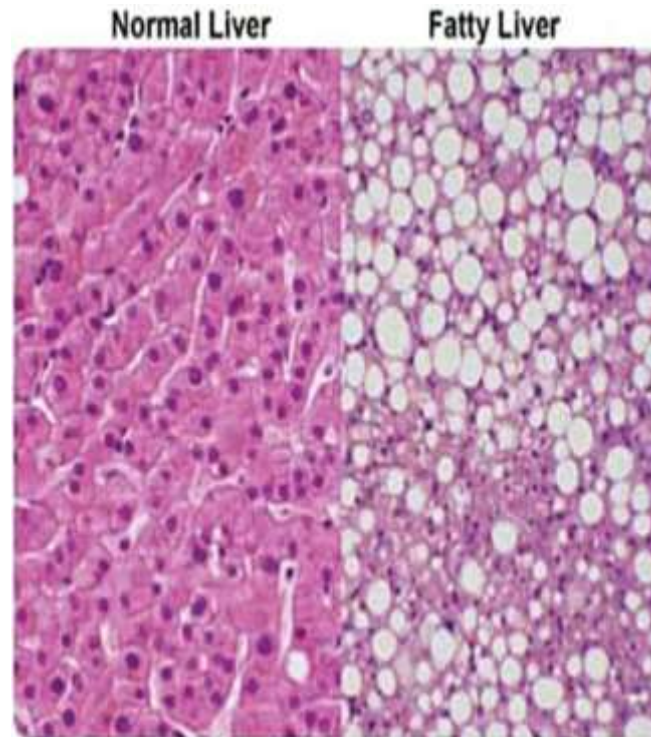
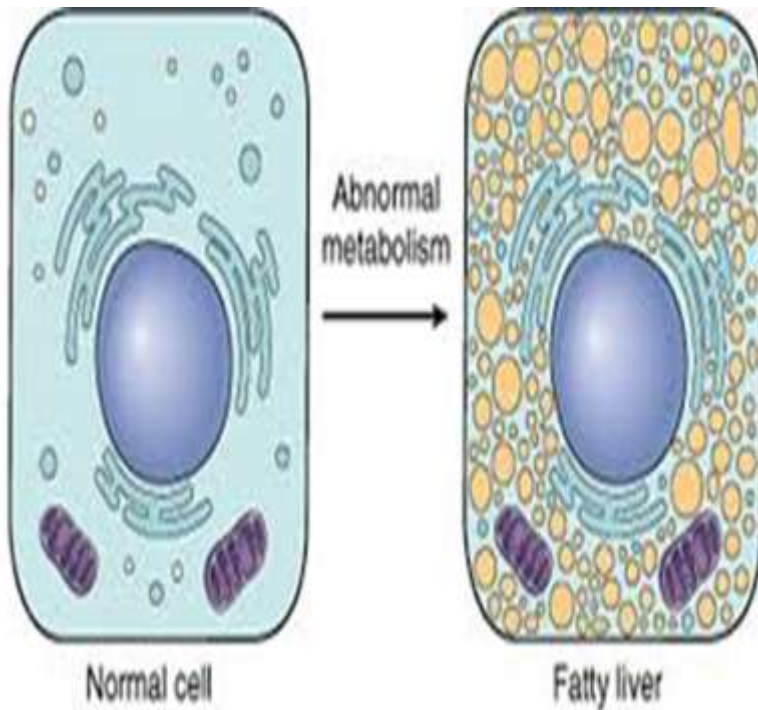
Categories of Intracellular Accumulations

1-Lipids

A-Triglycerides

Steatosis (Fatty Change)

Liver (most common), heart, kidney.



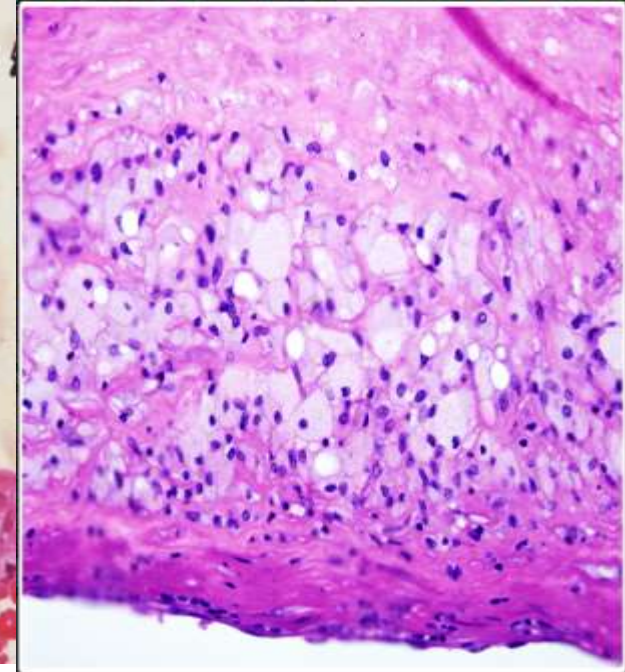
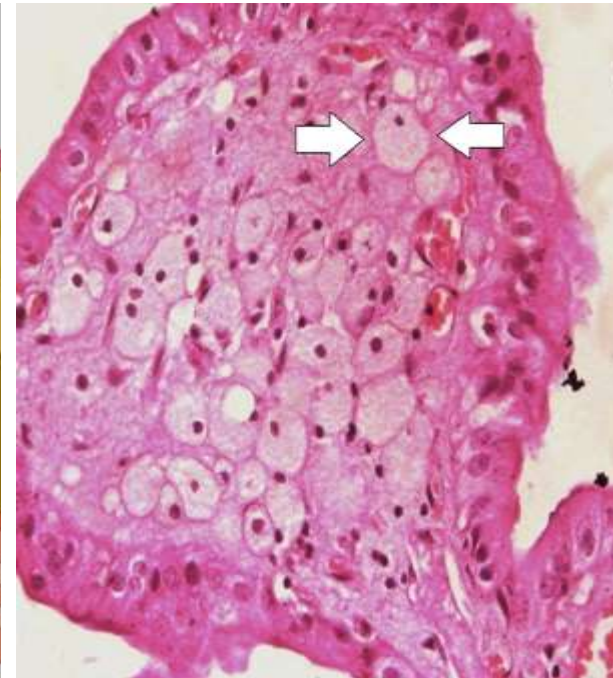
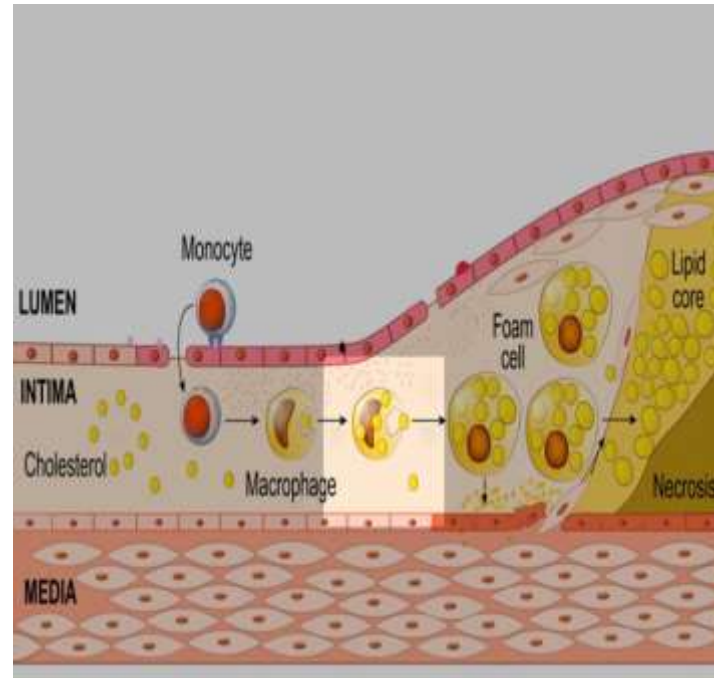
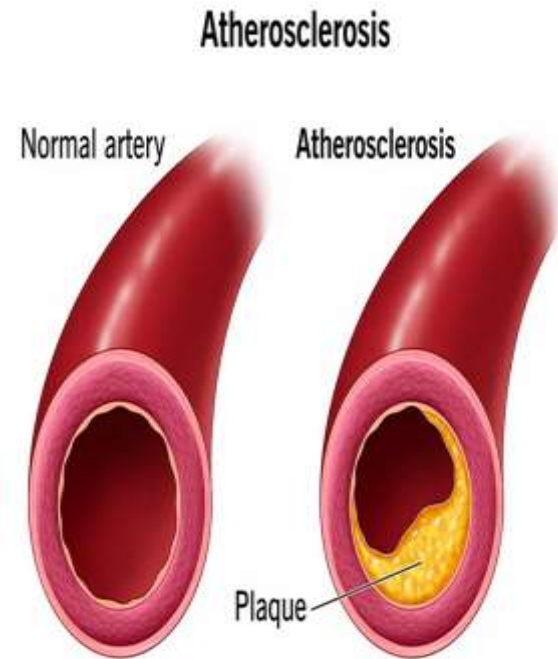
Categories of Intracellular Accumulations

1-Lipids

B-Cholesterol & Cholesteryl Esters

Atherosclerosis, [Xanthomas Xanthelasma]

arteries / [skin, tendons]



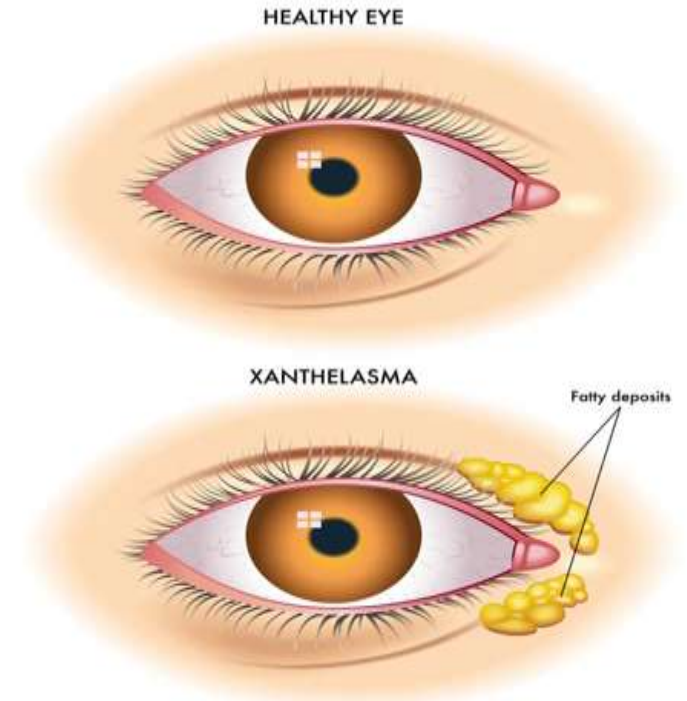
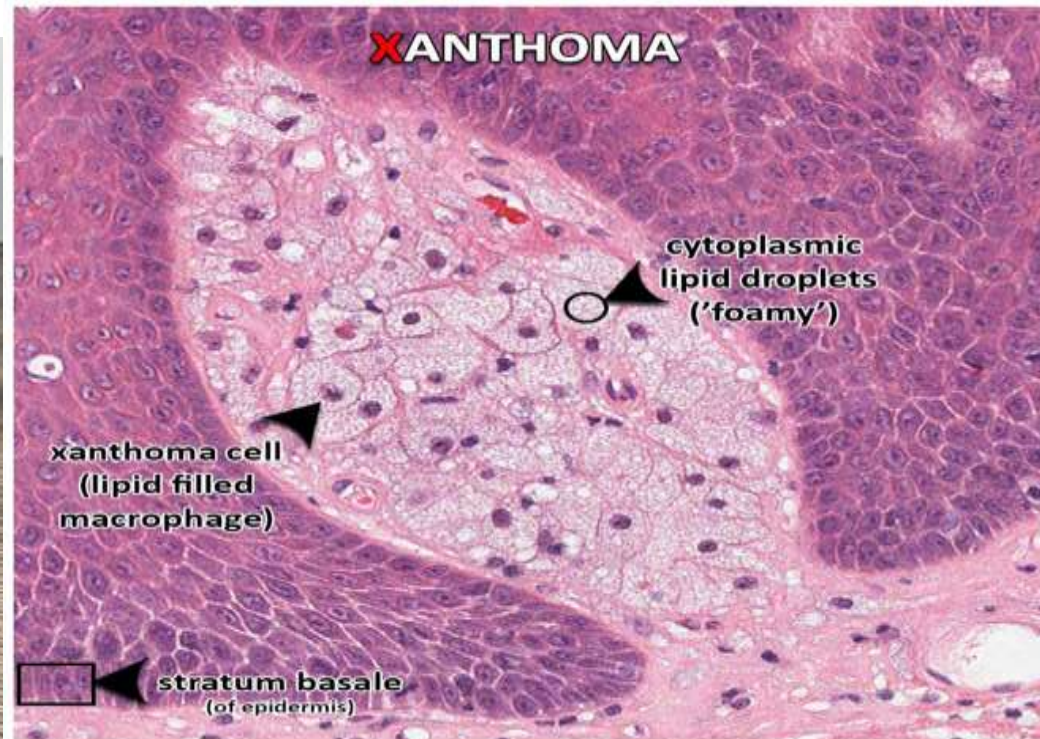
Categories of Intracellular Accumulations

1-Lipids

B-Cholesterol & Cholesteryl Esters

Atherosclerosis, Xanthomas Xanthelasma

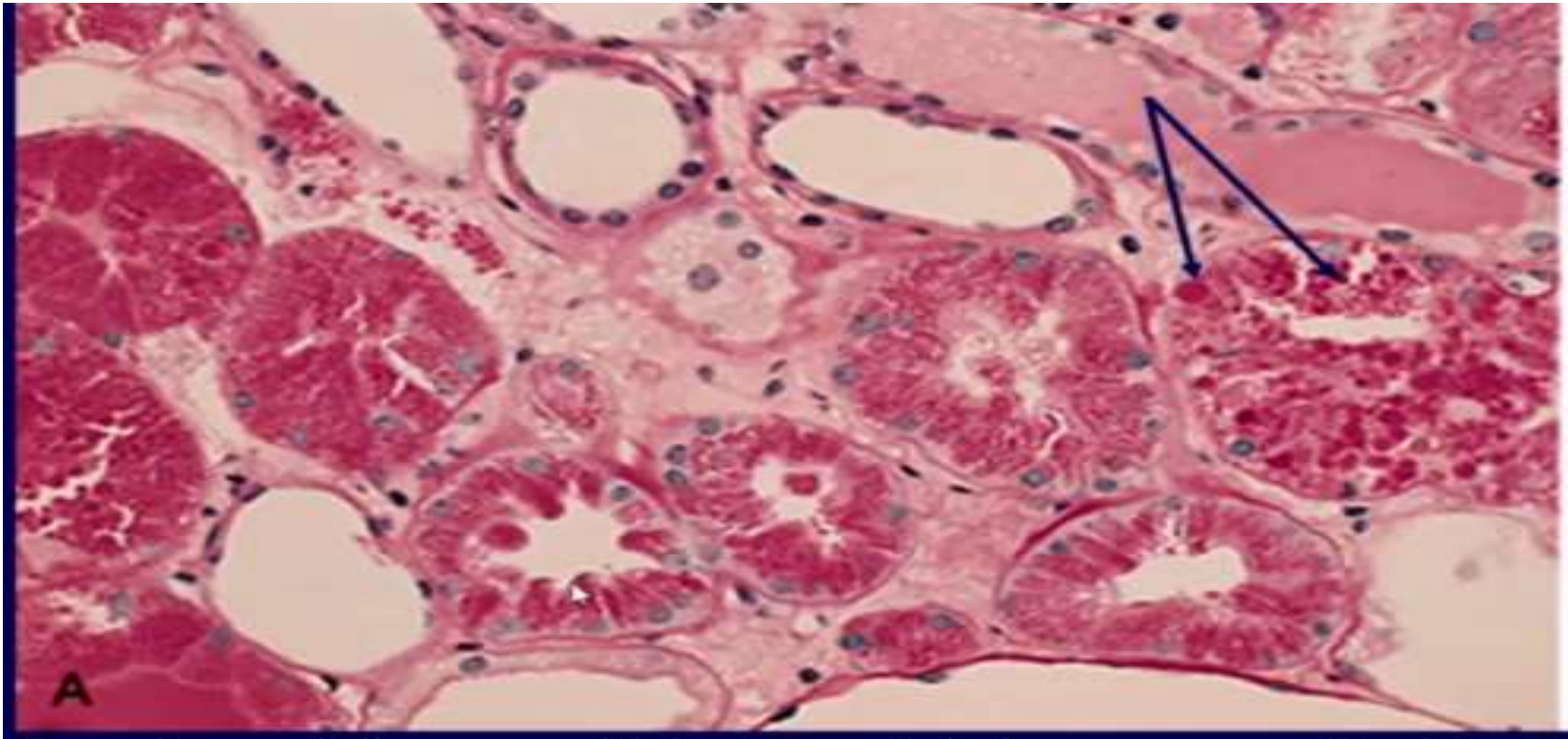
arteries / [skin, tendons]



Categories of Intracellular Accumulations

2-Proteins

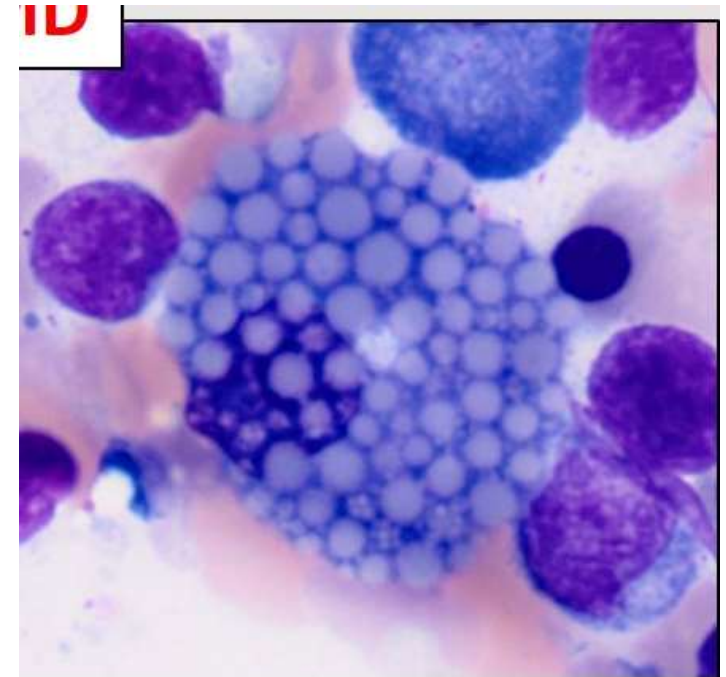
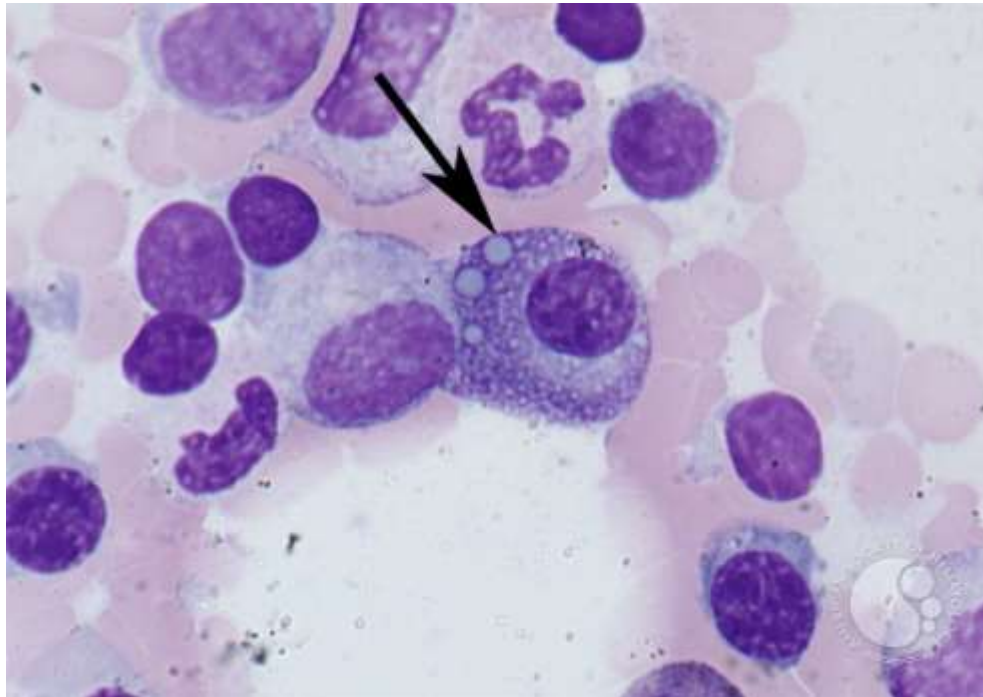
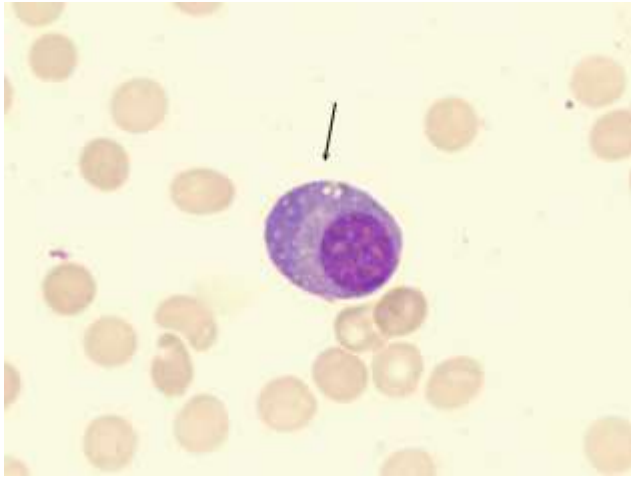
A-Proteinuria: Increased protein reabsorption by the proximal renal tubules in kidney disease (e.g., nephrotic syndrome).



Categories of Intracellular Accumulations

2-Proteins

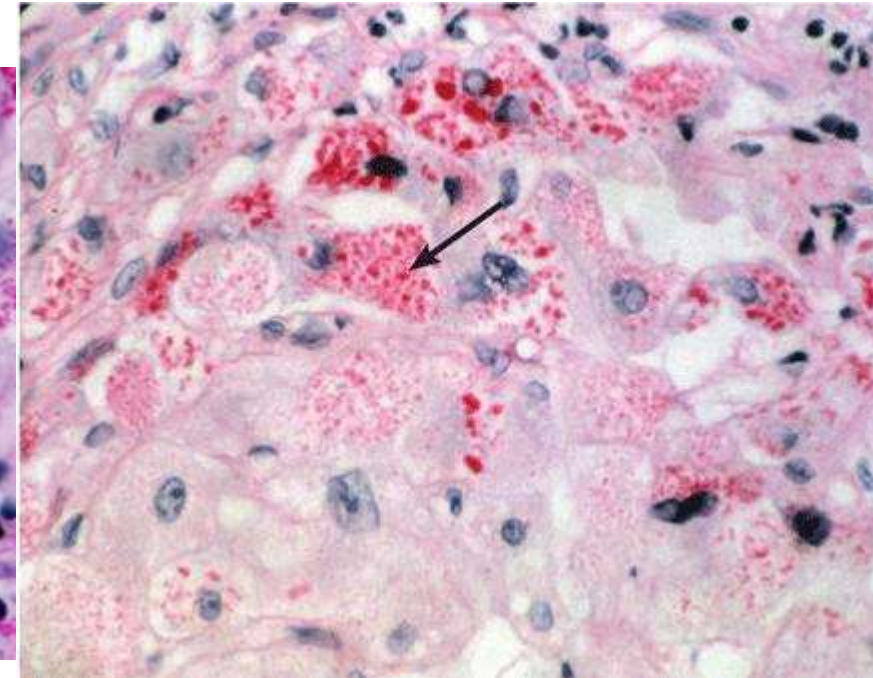
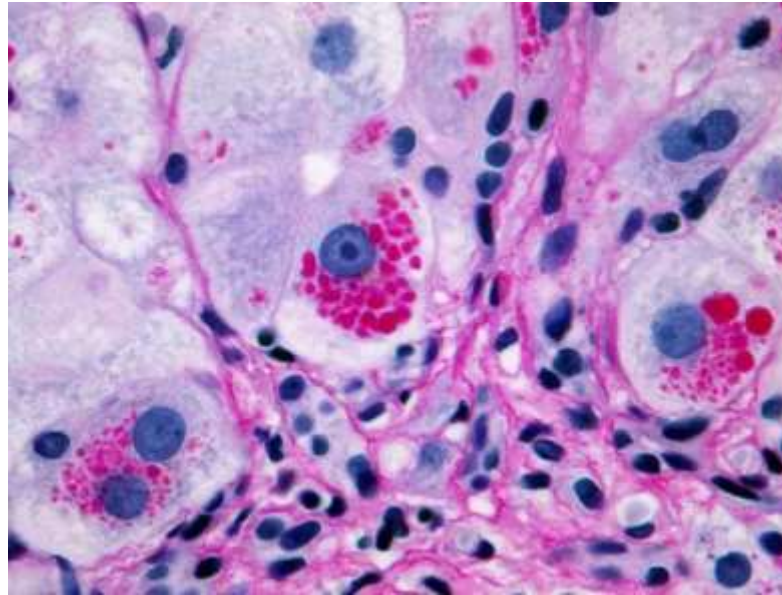
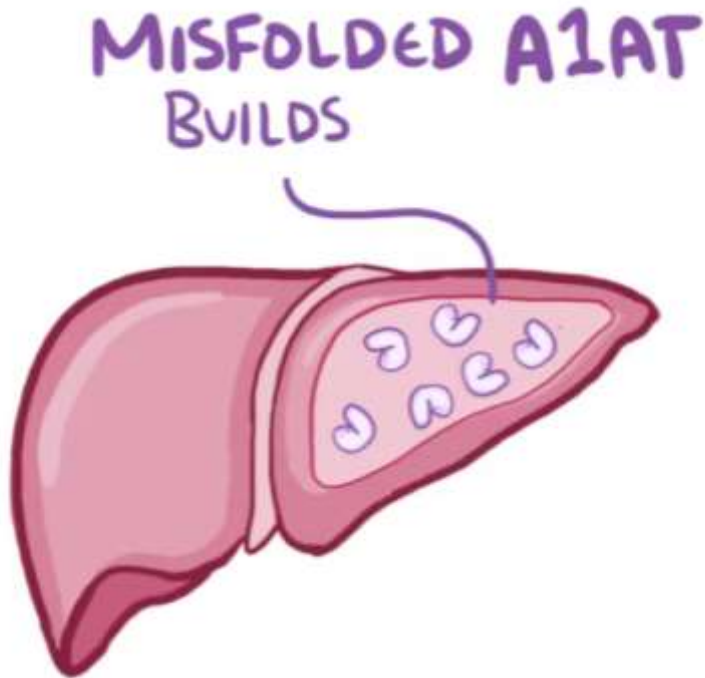
B-Plasma Cell Dyscrasia: Accumulation of newly synthesized, excessive immunoglobulins in the rough endoplasmic reticulum (RER) of plasma cells



Categories of Intracellular Accumulations

2-Proteins

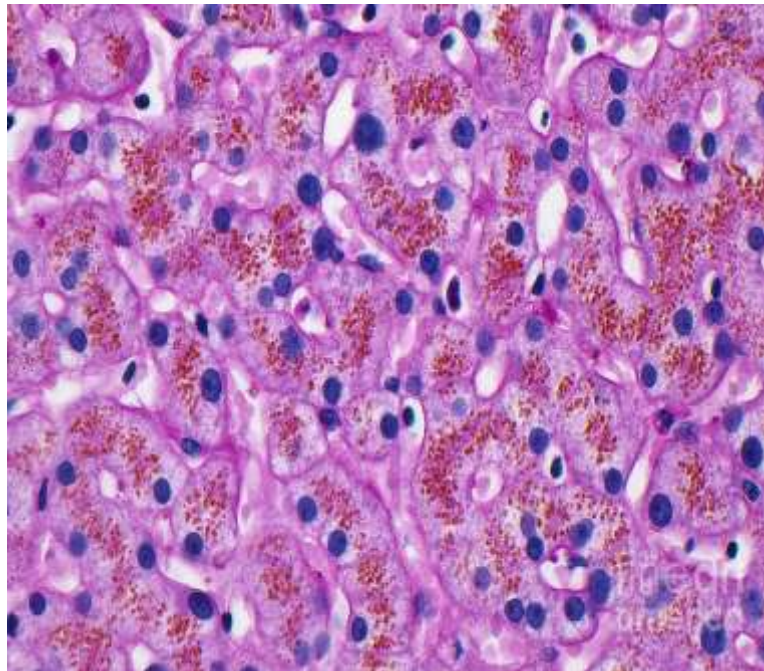
C-Alpha-1 Antitrypsin Deficiency: Defect in protein folding causes (A1AT) to accumulate in the **RER** of **hepatocytes** instead of being secreted.



Categories of Intracellular Accumulations

3-Pigments

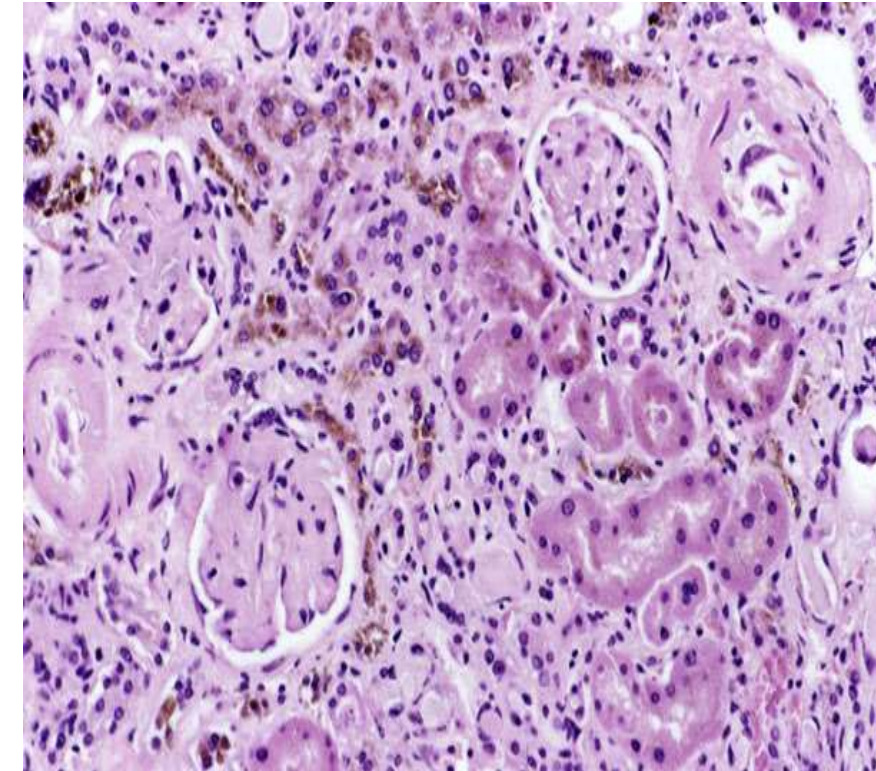
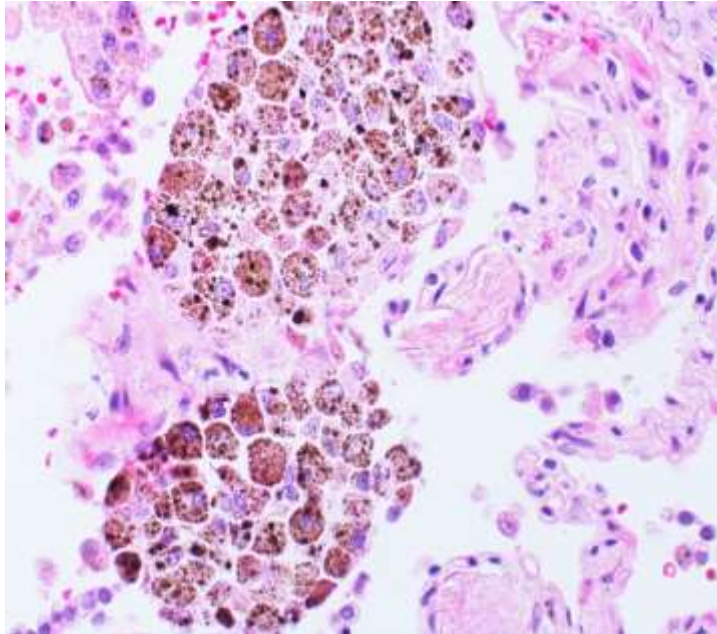
A-Lipofuscin: "Aging" pigment. (Indigestible residue remaining in the lysosome after incomplete breakdown of a normal or abnormal substance)



Categories of Intracellular Accumulations

3-Pigments

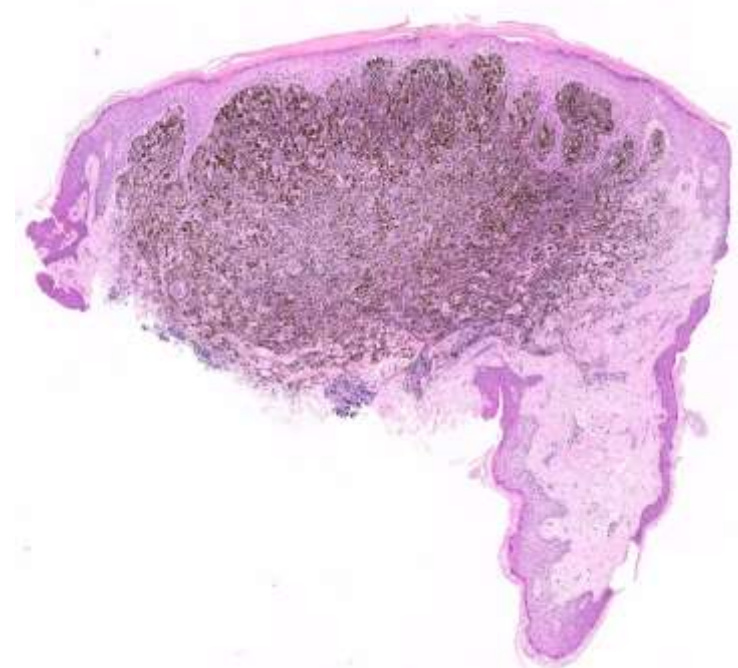
B-Hemosiderin : (Phagocytosis of an endogenous substance (red blood cells/hemoglobin) with subsequent accumulation of a non-degradable residue (iron)).



Categories of Intracellular Accumulations

3-Pigments

C-Melanin : (The accumulation is generally due to a localized increase in its normal synthetic process (e.g., tanning) or a focal proliferation of the melanocytes themselves (e.g., in a mole or melanoma).



Categories of Intracellular Accumulations

3-Pigments

D-Carbon/Coal Dust : Inhaled carbon particles, Phagocytosed by alveolar macrophages.

Healthy lung



Simple black lung disease



Complicated black lung disease

